SPACE · LINK · 1 · GROUND · 2 · PRODUCT · 3

TSYP 14 · TECHNICAL CHALLENGE · INFO SESSION

Ground Truth Under Attack

Securing the Earth Observation Chain

IEEE Computational Intelligence Society Tunisia Chapter

IEEE GRSS Geoscience and Remote Sensing Society

IEEE Signal Processing Society TUNISIA CHAPTER

Info Session — 20 September 2026

What We'll Cover



Why this challenge exists

01

The problem it solves, and the goal



The three tracks

02

Pick one ground link, sensor data, or imagery



Deliverables & phases

03

What every team submits, and when



Timeline & scoring

04

Key dates and how points are awarded



Prizes & how to join

05

1500 DT prize pool, eligibility, contact

AT A GLANCE

TEAMS OF UP TO

5 members

PHASE 1 DEADLINE

03 Nov 2026

FINAL DEADLINE

11 Dec 2026

PRIZE POOL

1500 DT

FINALISTS

up to 7 teams

Why Earth-Observation Trust Matters

Earth-observation products support decisions in agriculture, insurance, water management and disaster response.

Their trustworthiness depends on evidence of origin, time, integrity and processing history across the sensing, communication and product chain.

Teams select one segment ground link, in-situ sensing, or imagery and build a verifiable defensive prototype.



GRSS

Remote sensing



SPS

Signal / image processing



CIS

Computational intelligence & security analytics

A Failure That Stays Silent

Documented navigation interference, false-data injection, and increasingly capable image synthesis show that plausible-looking data may still be untrustworthy.



A failure may remain silent: the pipeline keeps operating while a corrupted observation reaches the analyst.

The challenge focuses on:

- **Provenance**
- **Integrity verification**
- **Anomaly detection**
- **Controlled adversarial testing**

CHALLENGE GOAL

Pick a Track. Defend It.

- Use authentic public data
- Apply the common standard scenario
- Detect or prevent the attack
- Report detection quality, false alarms, latency, memory and security overhead

ENTRY BARRIER



No purchase is required.

A software or emulated trust anchor and local laptop execution are fully valid. Optional hardware earns no scoring advantage.

One Chain Three Ways In

Each team selects exactly one track. All tracks share the same workload, metrics and scoring rubric.



TRACK 1

Secure Ground-Link Testbed

Can a stored signal or sequence be authenticated?

TRACK 2

False Data Injection

Is this sensor stream consistent with its context?

TRACK 3

Trust the Pixel

Was this image altered or falsely attributed?

TRACK 1

Secure Ground Link Testbed

Can a stored signal or sequence be authenticated?

Use recorded SatNOGS observations or organizer-supplied telemetry. Authenticate a local command/telemetry sequence and detect replay, sequence injection and digitally added interference. No live downlink is mandatory.

PROCESSING

Demodulated features, spectral occupancy, sequence analysis.

AI / DETECTION

Interference / anomaly classifier with calibrated confidence.

SAFETY

Files or conducted / coaxial / shielded setup only no radiated attack.

OFFICIAL REFERENCE DATA

SatNOGS Network Observations

STANDARD TEST

Replay a valid 5–20-frame subsequence, inject one sequence anomaly, and add narrowband interference digitally at a hidden SNR. The jury supplies a held-out recording.

TRACK 2

False Data Injection

Is this sensor stream consistent with its context?

Use a real environmental time series and public orbital / contextual observations. Detect a step bias, gradual drift and replay while accounting for spatial/temporal mismatch and natural variability.

PROCESSING

Time series, change points, sensor / context fusion.

AI / DETECTION

Compromise probability or gradual trust score.

OUTPUT

Alert, quarantine, verification request, or de weighting.

OFFICIAL REFERENCE DATA

NASA POWER Daily API + Copernicus Sentinel-2 L2A

STANDARD TEST

Inject a 12% step bias, a gradual drift up to 15%, and a seven-sample replay into a real series. Validate against contextual / orbital indicators. Site and interval are hidden in the final test.

TRACK 3

Trust the Pixel

Was this image altered or falsely attributed?

Bind an authentic EO image to a provenance record, then identify resampling / splicing, copy-move, or generated / inpainted content. Distinguish post-capture integrity from semantic truth.

PROCESSING

Frequency, resampling and forensic features.

AI / DETECTION

Detector evaluated on authentic EO imagery.

TRUST

Digest, time/source claim, and verification chain.

OFFICIAL REFERENCE DATA

EuroSAT (Sentinel-2)

STANDARD TEST

Authenticate clean patches, then detect a resampled/spliced image, a copy-move alteration, and a generated/inpainted region. The jury supplies held-out authentic and manipulated patches.

WHAT EVERY TEAM SUBMITS

Six Deliverables, One Track

1 **Threat model & segment analysis**

2 **Platform / architecture / trust anchor**

3 **Signal / image processing**

4 **AI or statistical detection**

5 **Adversarial validation & response**

6 **Documentation using the official reference datasets**

TWO PHASES

Anonymous First. Live Second.

PHASE 1 — ANONYMOUS SELECTION

Deadline: 03 November 2026

- Anonymous report + code submission
- No names, unit names, logos, or identifying repository metadata
- Administrative committee assigns a team code and sanitizes the repository before it reaches the jury
- Every registered team receives a Phase 1 score

PHASE 2 — PROTOTYPE & PITCH

Deadline: 11 December 2026

- Up to 7 finalists build a working prototype
- Live or replayable attack demonstration during TSYP 14
- Non-anonymous — the live pitch requires it
- 7-minute pitch + 3-minute jury Q&A, English

KEY DATES

Timeline



All Phase 1 deadlines are 23:59 GMT+1 · Up to 7 finalists advance to Phase 2 · TSYP 14 final-presentation date confirmed closer to the event

SCORING SYSTEM

100 Points, Two Phases

Final Score = 30% × Phase 1 + 70% × Phase 2

PHASE 1 · 30%

Threat model & relevance	15
Architecture & trust mechanism	15
Signal / image processing	15
Detector design & evaluation	20
Standard scenario & evasion	20
Feasibility & local execution	10
Clarity & reproducibility	5

PHASE 2 · 70%

End-to-end functional prototype	20
Provenance / security control	15
Processing & local detection	15
Held-out standard attack & evasion	20
Metrics, efficiency & accessibility	10
Documentation & reproducibility	10
Pitch & technical questions	10

Tie-break: held-out attack score → lower false-alarm rate → documented jury decision. No bonus points: society, YP or WIE membership does not affect the score.

RECOGNITION

1500 DT Prize Pool

3 winning teams · Total pool 1500 DT · Certificates for all finalists



Who Can Join, What's Required

- Student Branches (SB) only
- Hosted collaboratively by 3 IEEE chapters
- Teams of up to 5 members
- Exactly one track per team
- Official reference datasets required for the standard scenario
- Phase 1 anonymous · Phase 2 non-anonymous

EVALUATION

One Rubric, Every Team



Judged by an independent jury of IEEE academic representatives from the three co-organising chapters GRSS, SPS and CIS.

- The same jury serves both Phase 1 and Phase 2, for consistent scoring
- Jury members declare conflicts and recuse themselves from affected teams
- Every submission is judged against the same published rubric no exceptions

JOIN THE CHALLENGE

Let's Build Something Real

Register your team, choose your track, and start with the reference datasets. Questions before then? We're one email away.

CONTACT

maha.charfeddine.tn@ieee.org
boudour.ammar@ieee.org
ghassen.ltaif@ieee.org

NEXT DEADLINE

Phase 1 — 03 November 2026

IEEE GRSS · SPS · CIS

Tunisia Section Chapters

Thank you.

Questions?